NEWTON CLASSES

Class 10 Maths — Quick Formula Bank

Starter revision resource • Verify the current Bihar Board syllabus with your school.

Real Numbers

• For positive integers a and b: HCF(a,b) × LCM(a,b) = a × b.

• Euclid's division lemma: a = bq + r, where 0 ≤ r < b.

Polynomials

• For $ax^2 + bx + c$ with zeroes α and β: $\alpha + \beta = -b/a$; $\alpha\beta = c/a$.

• Quadratic polynomial from zeroes α, β: $x^2 - (\alpha + \beta)x + \alpha\beta$ (monic form).

Pair of Linear Equations

• For a■x + b■y + c■ = 0 and a■x + b■y + c■ = 0: unique solution if a■/a■ ≠ b■/b■.

• No solution if a■/a■ = b■/b■ ≠ c■/c■; infinitely many if all three ratios are equal.

Quadratic Equations

• For $ax^2 + bx + c = 0$: $x = (-b \pm \sqrt{(b^2 - 4ac)})/(2a)$, $a \neq 0$.

• Discriminant $D = b^2 - 4ac$: $D > 0$ two distinct real roots; $D = 0$ equal real roots; $D < 0$ no real roots.

Arithmetic Progressions

• nth term: a■ = a + (n – 1)d.

• Sum of first n terms: S■ = n/2 [2a + (n – 1)d] = n/2(a + l).

Coordinate Geometry

• Distance between (x■,y■) and (x■,y■): $\sqrt{[(x■-x■)^2 + (y■-y■)^2]}$.

• Midpoint: ((x■+x■)/2, (y■+y■)/2).

• Section formula, internal ratio m:n: ((mx■+nx■)/(m+n), (my■+ny■)/(m+n)).

Trigonometry

• sin θ = perpendicular/hypotenuse; cos θ = base/hypotenuse; tan θ = perpendicular/base.

• $\sin^2\theta + \cos^2\theta = 1$; $1 + \tan^2\theta = \sec^2\theta$; $1 + \cot^2\theta = \operatorname{cosec}^2\theta$.

• For complementary angles: sin(90°–θ)=cos θ; tan(90°–θ)=cot θ.

Circles and Areas

• Circumference = $2\pi r$; area of circle = πr^2.

• Arc length = (θ/360°) × $2\pi r$; sector area = (θ/360°) × πr^2.

• Area of sector minus triangle gives minor segment area when the central angle is θ.

Surface Areas and Volumes

• Cylinder: curved surface $2\pi rh$; total surface $2\pi r(r+h)$; volume $\pi r^2 h$.

• Cone: slant height $l = \sqrt{(r^2+h^2)}$; curved surface πrl; volume $(1/3)\pi r^2 h$.

• Sphere: surface area $4\pi r^2$; volume $(4/3)\pi r^3$; hemisphere volume $(2/3)\pi r^3$.

Statistics and Probability

• Mean for grouped data: $\bar{x} = \Sigma f x / \Sigma f$.

• Probability of event E = favourable outcomes / total equally likely outcomes.

• $0 \leq P(E) \leq 1$; $P(\text{not } E) = 1 - P(E)$.

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